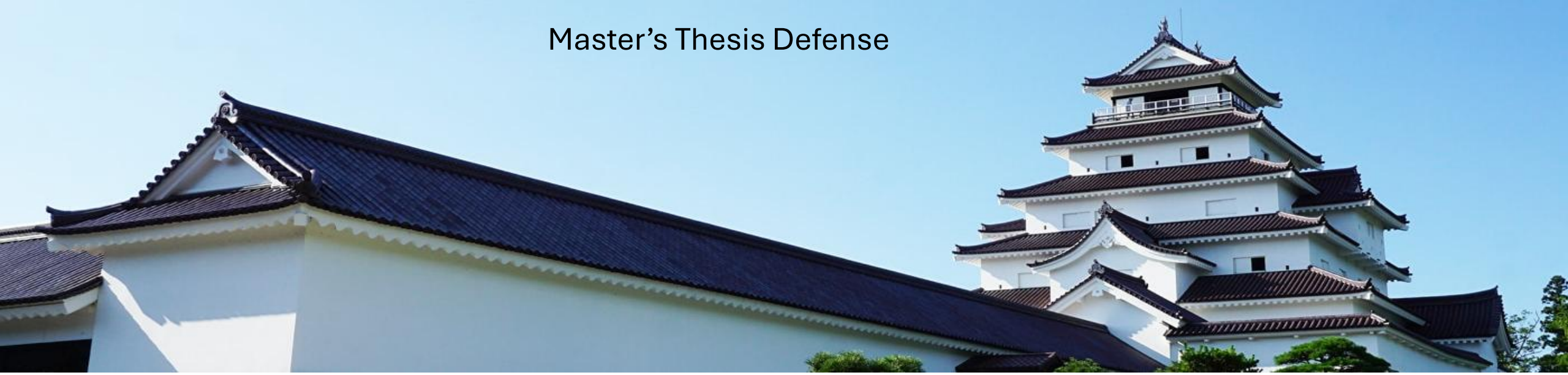
Master's Thesis Defense



Urgency-Aware QoE-Driven Multi-UAV Fleet Trajectory Optimization for Multi-Priority Post-Disaster Networks via MADRL

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Outline

1. Introduction & Motivation
2. Problem Formulation
3. Proposed K-MAPPO Framework
4. Performance Evaluation
5. Conclusion

Motivation: Post-Disaster UAV-Assisted Communication

The Problem:

Terrestrial networks are completely destroyed during disasters, creating critical connectivity gaps.

UAV-Assisted Communication:

- Unmanned Aerial Vehicles (UAVs) act as temporary aerial base stations.
- High mobility, rapid deployment, and line-of-sight (LoS) altitude benefits.



Disaster



Infrastructure Collapse



UAV Deployment

Figure sources:
https://share.google/1Fu2Lo93rsA3lmXoF
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Primary Challenges for UAV Trajectory Design

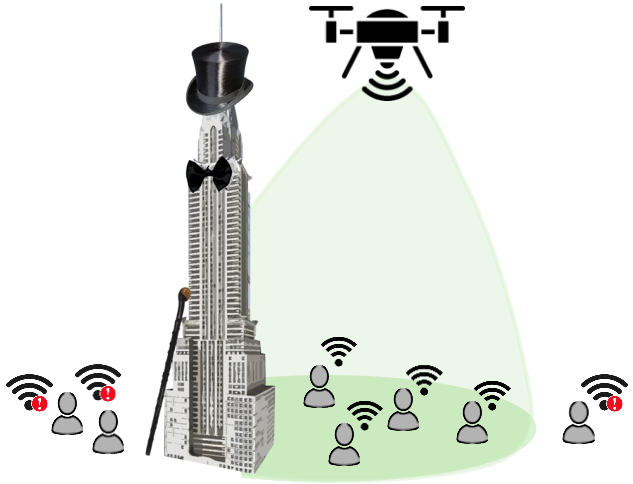
Limited Coverage of Single-UAV Systems

- A single UAV has limited coverage to service in large scale post-disaster areas.
- Need a real-time, adaptive, multi-UAV trajectory control.



Complex 3D Urban Environment

- Dynamic user mobility.
- Signal degradation due to 3D physical blockages.
- Altitude-coupled beamwidth trade-offs.



Heterogeneous User Urgency

- Disaster zones contain asymmetric user priorities (Emergency vs. Normal).
- Static paths cannot handle dynamic emergency event arrivals.

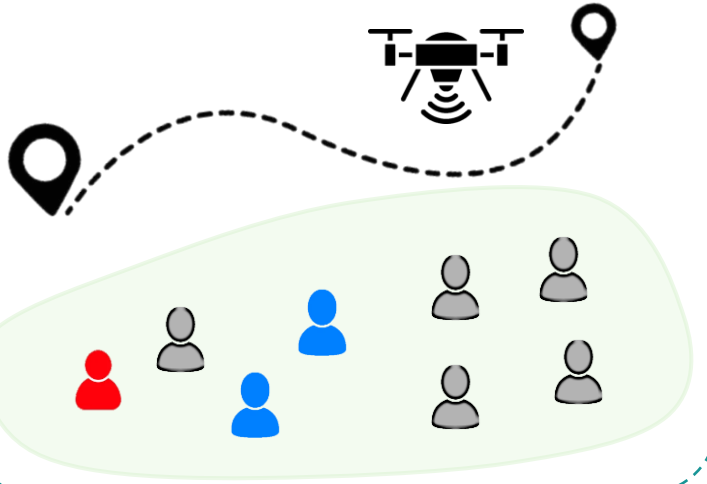


Figure sources:
https://share.google/Tig5RYrHxEsBR3QQu
https://share.google/QS9Fh2iXmDqJvj2ch

Conceptual Framework: QoE & Urgency-Aware Utility

Subjective QoE Modeling via Mean Opinion Score (MOS)

Objective: Maximize user satisfaction via demand-centric resource management [8].

1. **Logarithmic Mapping (Continuous score x):** The user data rate $R_{n,t}$ is mapped to a continuous quality score $x_{n,t}$

$$x_{n,t} = a \cdot \log_{10}\left(\frac{R_{n,t}}{b}\right) \qquad a = \frac{x_2 - x_1}{\log_{10}(\theta_2/\theta_1)} \qquad b = \frac{\theta_1}{10^{x_1/a}}$$

- Range: $\theta_1 = 0.5$ Mbps (MOS 1) to $\theta_2 = 5.0$ Mbps (Saturation, MOS 5)

2. **Discrete Mapping to MOS:** MOS is defined on a 1–5 scale to reflect human-perceived quality:

$$MOS_{n,t} = \begin{cases} 0, & R_{n,t} \le R_{th} \\ 1, & R_{n,t} > R_{th} \text{ and } x_{n,t} < 1.5 \\ 2, & 1.5 \le x_{n,t} < 2.5 \\ 3, & 2.5 \le x_{n,t} < 3.5 \\ 4, & 3.5 \le x_{n,t} < 4.5 \\ 5, & x_{n,t} \ge 4.5 \end{cases}$$

- Outage (MOS = 0): $R_{n,t} \le 100$ kbps (Hardware limit).
- Discrete Scale: 1 (Poor) to 5 (Excellent) based on $x_{n,t}$ thresholds.

Urgency-Embedded MOS (Connectivity Relief Effect)

The Limitation: Standard MOS benchmarks network QoE uniformly, failing to differentiate urgency across heterogeneous user classes.

Concept: A low-throughput connection can deliver critical relief to an emergency user, but is of little value to a standard user.

Introduced Priority Factor $(\alpha_{n,t})$:

$$\alpha_{n,t} \in \{\alpha_{\text{Normal}}, \alpha_{\text{Important}}, \alpha_{\text{Emergency}}\},$$

In this work, $\alpha_{n,t} = \{1.0, 1.5, 2.0\}$ respectively.

- The priority-aware satisfaction $S_{n,t}$ embeds this urgency directly into the standardized MOS metric:

$$S_{n,t} = \min(\alpha_{n,t} \text{MOS}_{n,t}, 5)$$

Note: Priority weights (1.0, 1.5, 2.0) **are open input parameters**. The framework supports empirical survey data directly without changing the underlying optimization model.

[8] H. Dai, Y. Ju, Y. Liang, Z. Zhang, H. Xu, and B. Wang, "QoE-driven multi-UAV deployment scheme for emergency communication networks," Electronics Letters, vol. 60, no. 1, pp. 1–10, Jan. 2024.

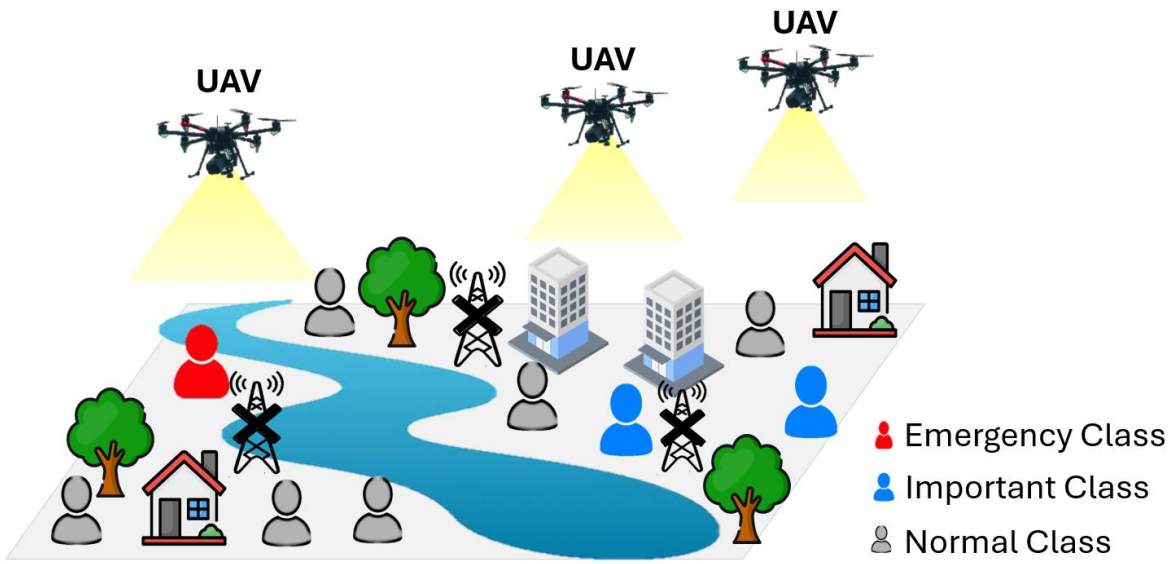
Research Goal & Proposed Approach

Objective: To design an autonomous, real-time 3D trajectory optimization framework for a multi-UAV fleet that maximizes priority-aware ground user Quality of Experience (QoE) in a post-disaster urban environment.

Proposed Approach:

1. **Autonomous Real-Time Trajectory Optimization:**
   - Train a multi-UAV fleet to learn optimal 3D trajectories in real-time in complex 3D urban environments.

2. **Priority-Aware QoE Framework:**
   - Utilize an urgency-aware utility framework to dynamically prioritize critical ground users.
   - Maximize priority-weighted user Quality of Experience (QoE) over raw throughput.



Targeted Area

Research Gap

Comparative analysis of existing UAV trajectory works

Reference	3D Env.	Multi-UAV	RL/ MARL	QoE/ MOS	Priority-Aware
Chowdhury et al. [1]					
Cherif et al. [2]	✓				
Hoang et al. [3]			✓		
Hoang et al. [4]		✓	✓		
Ruan et al. [5]		✓			
Guan et al. [6]		✓	✓		
Hoang et al. [7]	✓	✓	✓	✓	
Dai et al. [8]		✓		✓	
Wan et al. [9]	✓	✓	✓		
Xue et al. [10]	✓		✓		
Pan et al. [11]	✓	✓	✓		
Hoseini et al. [12]			✓		✓
Hu et al. [13]					✓
Our Work (Phase 1) [14]			✓	✓	
Our Work (Phase 2) *			✓	✓	✓
This Thesis	✓	✓	✓	✓	✓

Limitations of Prior Work:

- Many studies neglect 3D urban terrain blockages in trajectory optimization.
- Heavy focus on network-centric, some consider a uniform QoE metric optimization.
- Existing rigid user classification models lack spatial priority tracking flexibility.

Our Contribution:

Bridges all gaps by pairing a 3D urban environment with a multi-priority-aware, QoE-driven MADRL trajectory framework.

*This work was accepted for publication in the proceedings of the IEEE International Conference on Consumer Electronics (ICCE 2026).

Abbreviations:
RL: Reinforcement Learning
MARL: Multi-Agent Reinforcement Learning
MADRL: Multi-Agent Deep Reinforcement Learning
QoE: Quality of Experience
MOS: Mean Opinion Score
3D Env.: Three-dimensional (3D) Environment

Outline

1. Introduction & Motivation
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Problem Formulation

1. **Objective Function:** to optimize the 3D trajectories of the UAV fleet to maximize the collective priority-weighted satisfaction of ground users over the mission duration.

$$\mathcal{P}: \max_{\{\mathbf{q}_{\mathrm{m,t}}\}} \sum_{t=1}^{T} \sum_{n=1}^{N} w_{n,t} S_{n,t}$$

$S_{n,t}$: Priority-scaled satisfaction score $\min\left(\alpha_{n,t}\mathrm{MOS}_{n,t}, 5\right)$

$w_{n,t}$: Optimization priority weight hierarchy ($w_{\mathrm{Emergency}} > w_{\mathrm{Important}} \gg w_{\mathrm{Normal}}$)

2. **Operational Constraints:**

- 3D Flight Boundary: $\mathbf{q}_{\mathrm{m,t}} \in [X_{\min}, X_{\max}] \times [Y_{\min}, Y_{\max}] \times [Z_{\min}, Z_{\max}], \forall m, t$
- Velocity Bound: $\| \mathbf{q}_{\mathrm{m,t+1}} - \mathbf{q}_{\mathrm{m,t}} \| \leq v_{\max} \Delta t, \ \forall m, t$
- Coupled Altitude-Beamwidth: $\theta_{m,t} = f\left(z_{m,t}\right), \Longrightarrow z \in [100m, 250m] \rightarrow \theta \in [60^{\circ}, 120^{\circ}]$
- Service Continuity: Enforces minimum duration thresholds ($T_{\min}^{c_n}$) to prevent user service starvation for high-priority nodes (Important & Emergency).

Outline

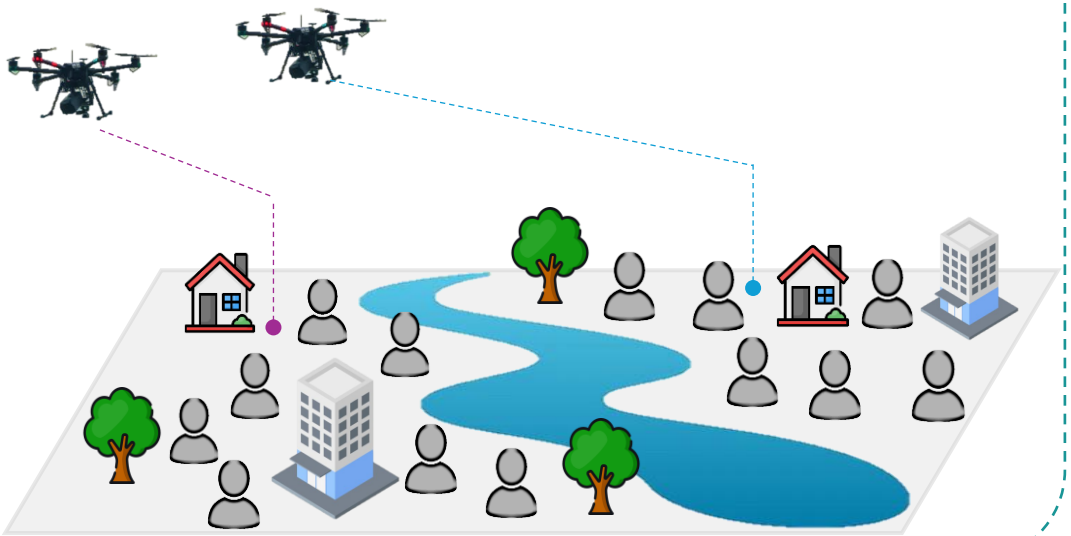
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Proposed K-MAPPO Framework Overview

To bypass early training inefficiencies and enhance policy convergence, we propose a two-stage operational design.

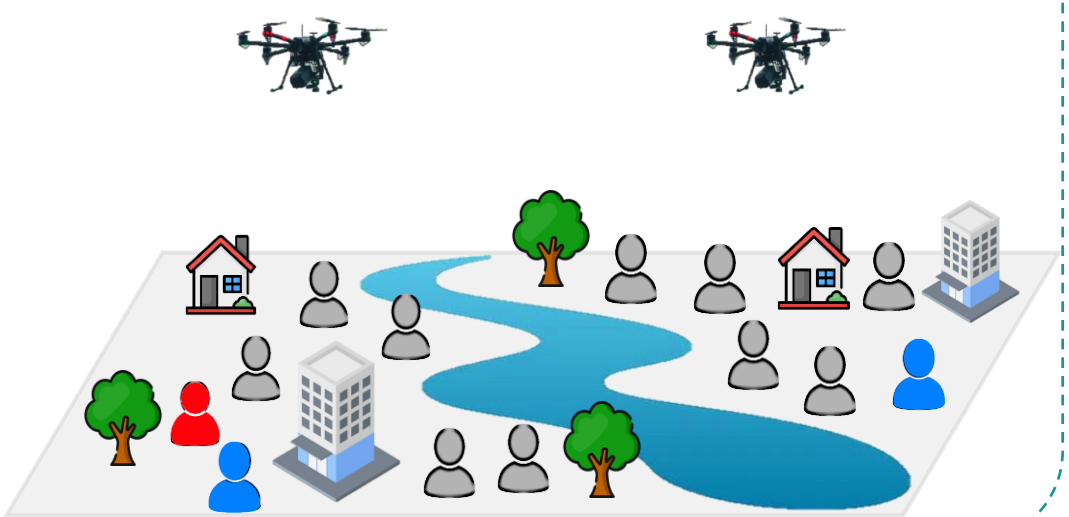
Stage I: Macro-Deployment (K-Means)

Computes ground user coordinates at episode reset to calculate optimal initial entry points for immediate network coverage.



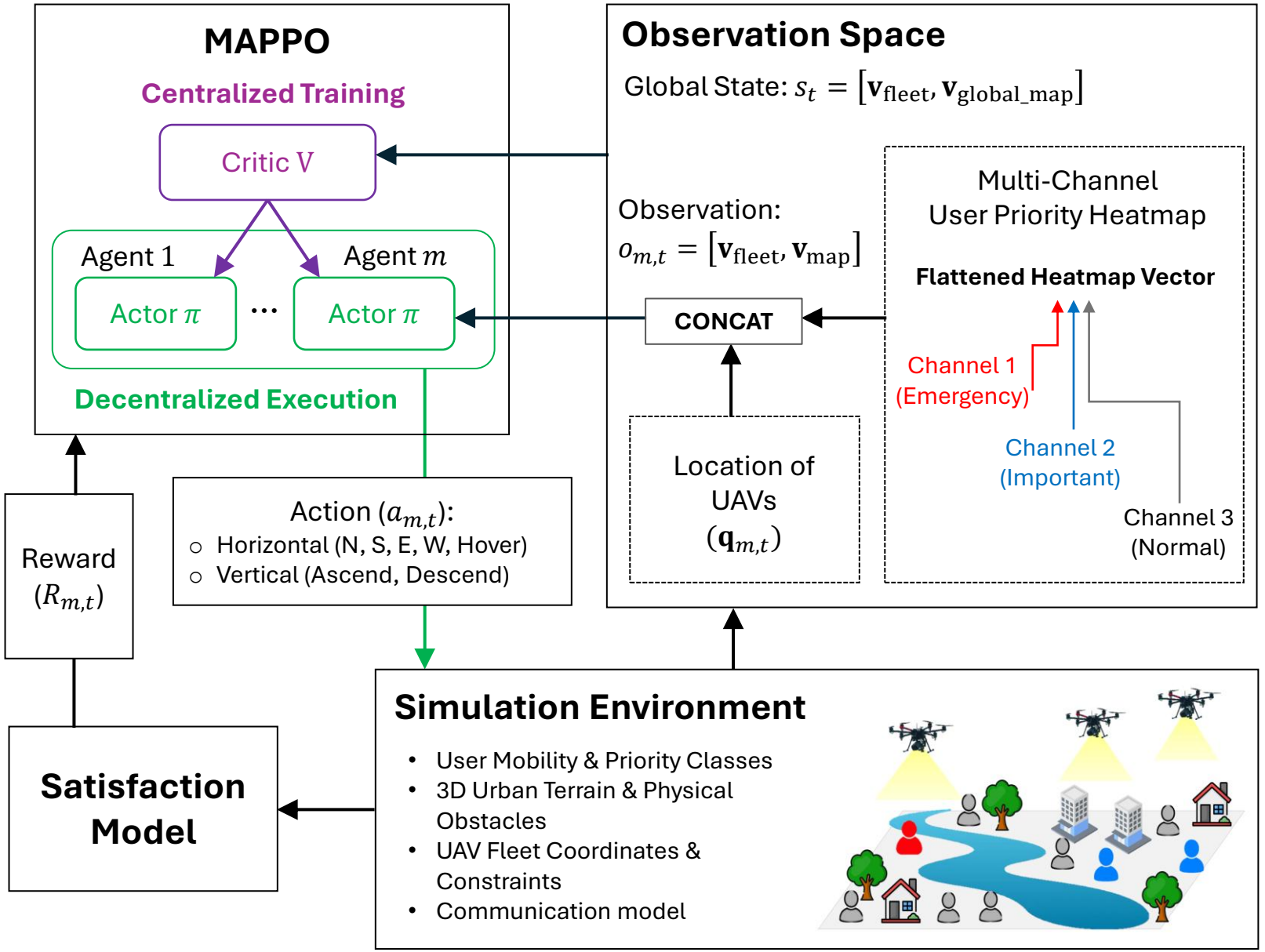
Stage II: Active Control (MAPPO)

Takes full autonomous control over the flight horizon, navigating dynamically to maximize priority-weighted user satisfaction.



MAPPO: Multi-Agent Proximal Policy Optimization

Proposed MADRL Training Loop — MAPPO-CTDE

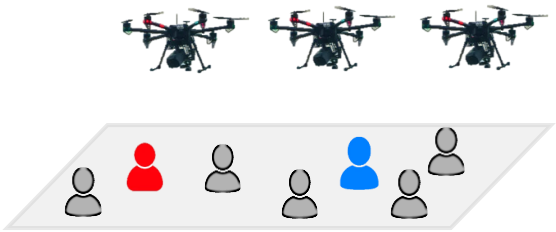


Curriculum Learning Strategy

A three-phase curriculum that progressively scales complexity to stabilize training.

Phase 1: Basic Navigation (Cooperative Flight)

Environment: Fixed altitude, free-space path loss (no blockages).
Objective: Learn basic horizontal tracking and cooperative anti-clustering.



Phase 2: 3D Pathfinding & Altitude-Coupled Coverage Trade-offs

Environment: Variable altitude $[z_{\min}, z_{\max}]$ with vertical actions.
Objective: Learn the altitude-coverage trade-off:

- High altitude: Wide footprint for scattered users.
- Low altitude: Narrow beam to maximize SNR and QoE.



Phase 3: Complete Urban Environment

Environment: 3D urban terrain with building blockages.
Objective: Learn to maintain LoS links and mitigate building shadows.



Reward Function Design

Composite Training Reward:

$$R_{m,t} = R_{\text{QoE},m,t} + R_{\text{guide},m,t} - P_{\text{delay},m,t} - P_{\text{overlap},m,t} - P_{\text{collision},m,t}$$

Weight Tuning Optimization:

$$\lambda_{\text{collision}} \gg \lambda_{\text{QoE}} > \lambda_{\text{overlap}} > \lambda_{\text{delay}} > \lambda_{\text{guide}}$$

Core Trajectory Mechanisms:

- **QoE Credit (R_{QoE}):** Distributed as 65% global fleet reward for cooperation, 35% local agent reward to avoid hotspot clustering.
- **Spatial Guidance (R_{guide}):** Auxiliary gradient pulling agents toward high-deficit user zones, suppressed within a 400m coordination radius.
- **Service Delay (P_{delay}):** Penalizes unserved time steps specifically for Emergency users to minimize wait times.
- **Fleet Spacing** ($P_{\text{overlap}}, P_{\text{collision}}$): Allowable 20% footprint handover buffer. Enforces a mandatory 50m physical UAV separation distance.

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Simulation Environment Settings

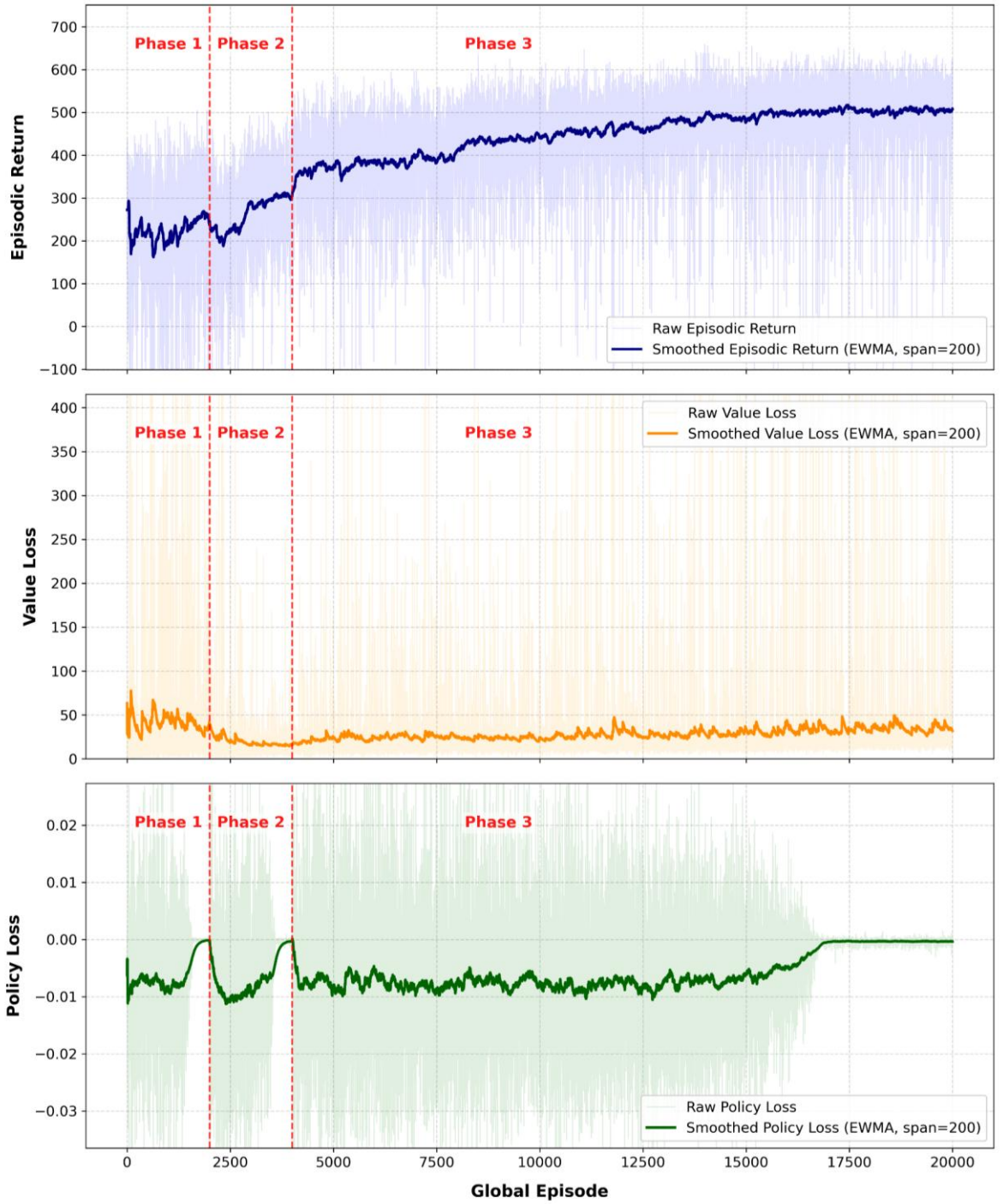
- **Simulation Setup:** $2000 \times 2000m$ 3D urban terrain with building blockages, servicing 150 mobile users ($v = 1.5m/s$).
- **Fleet Configuration:** $M = 3$ UAVs, max speed $v_{\max} = 25m/s$, operating altitude $z \in [100m, 250m]$, altitude-coupled beamwidth $\theta \in [60°, 120°]$.

Category	Parameter / Setup	Value / Configuration
Communication	Frequency / Bandwidth	5.8 GHz / 20 MHz (Synchronized ICIC)
	Transmit Power / Noise floor	25 dBm (UAV downlink) / −90 dBm thermal noise
Channel Model	Path loss / Shadowing	3D FSPL + NLoS shadowing, PL exp 3.2, Rician k=50
User Traffic	Distribution	150 users, hotspot spread $\sigma = 450m$
	Disaster wave schedule	Initial Normal → Important → Emergency → Mixed
Priority Mix & Weights	Tiers/α/w	Normal $(94\%, \alpha = 1.0, w = 0.1)$ Important $(4\%, \alpha = 1.5, w = 1.5)$ Emergency $(2\%, \alpha = 2.0, w = 3.0)$
Training Setup	Curriculum Training	K-Means + MAPPO CTDE over 20,000 total episodes

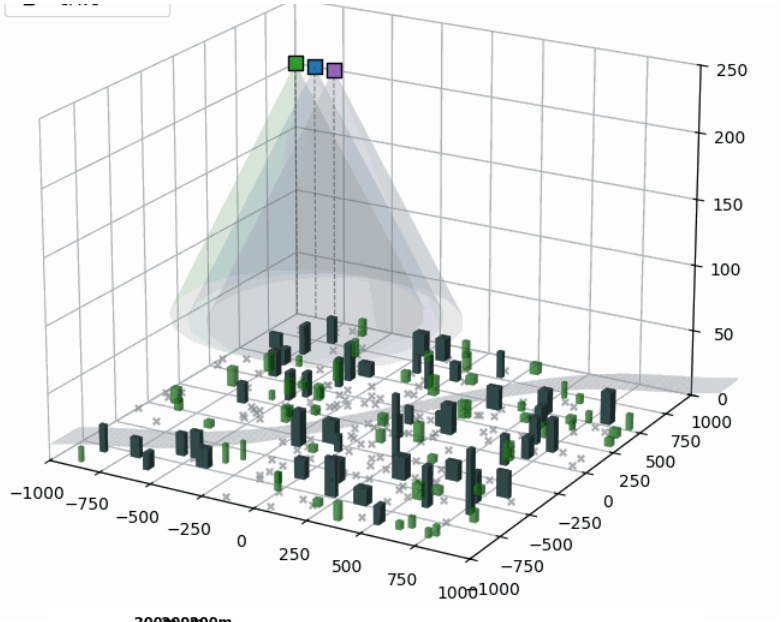
Results (1): Learning Performance

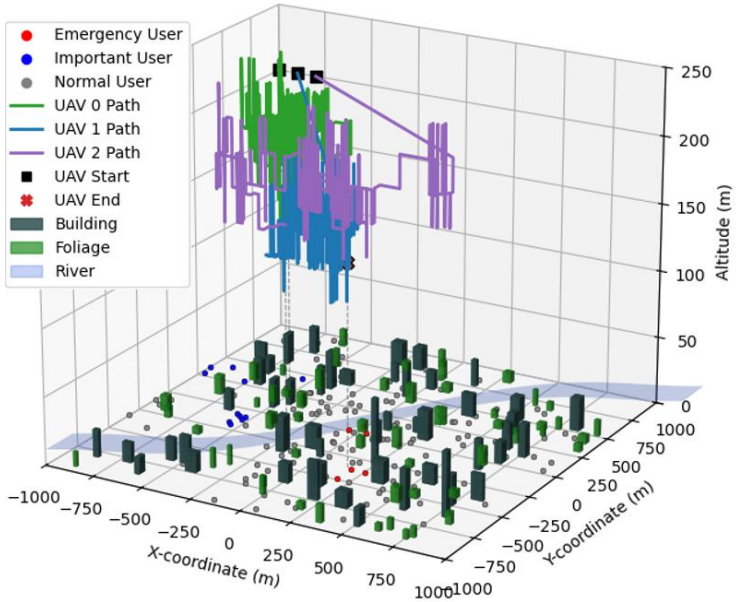
Progressive Learning Across Curriculum:

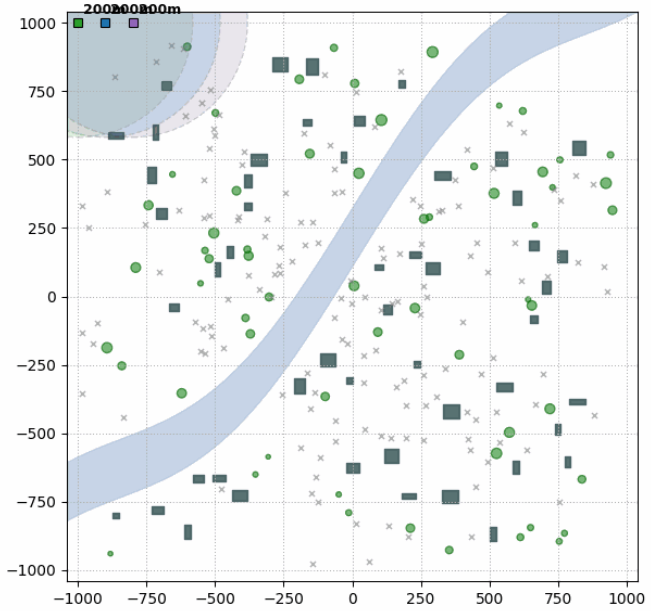
- **Episodic Return:** Trends consistently upward across Phase 1, 2, and 3, confirming policy learning.
- **Value Loss:** Remains slightly above zero due to sparse dynamic emergency wave arrivals, but maintains a stable baseline.
- **Policy Loss:** Converges smoothly near zero, demonstrating stable fleet trajectory convergence.

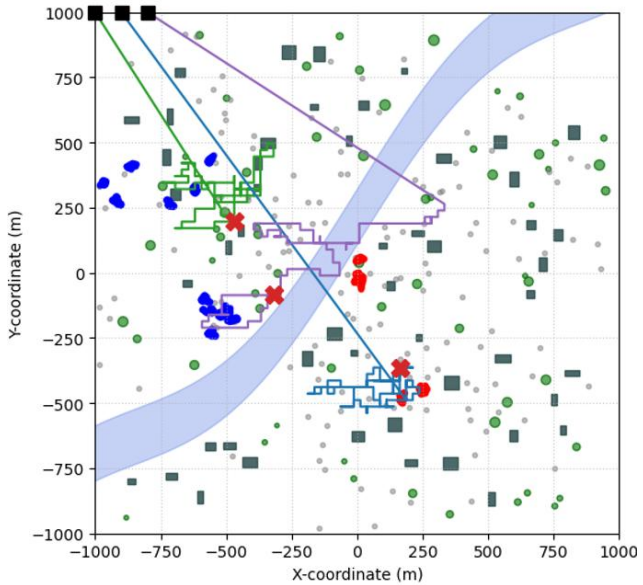


Results (2): Trajectory and Behavior









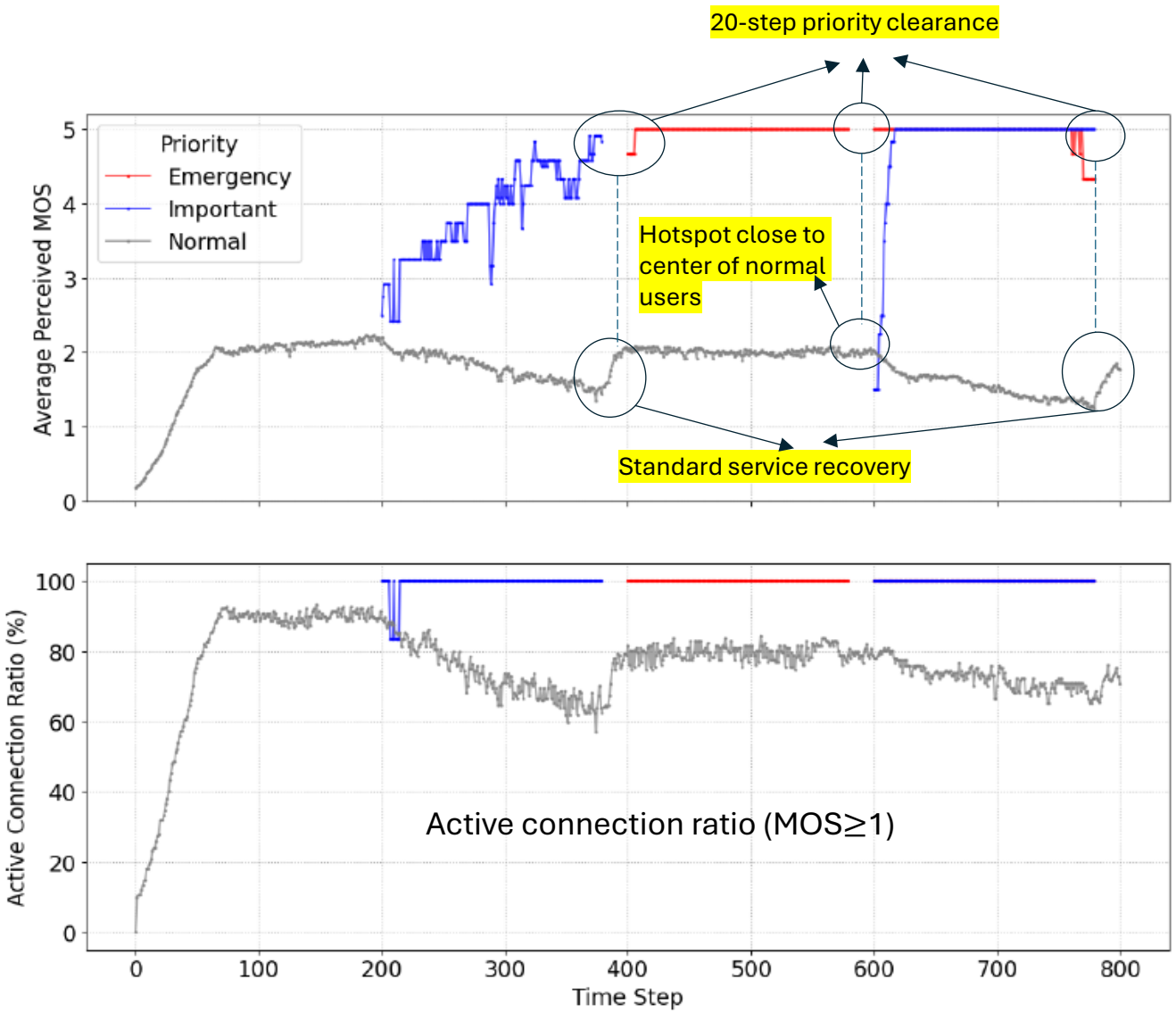
Two-Stage Hybrid Control in Action:

- **Stage I:** Straight-line flight trajectories from starting points to K-Means centers.
- **Stage II:** Dynamic MAPPO trajectory control takes over for real-time active tracking.

Coupled Altitude & Footprint Adaptation:

- **Ascending ($z \uparrow$):** Widens antenna footprint to search for disconnected users and mitigate building-induced signal blockages.
- **Descending ($z \downarrow$):** Narrows antenna beam, maximizing SNR and perceived QoE over high-priority users.

Results (3): Temporal Satisfaction



Dynamic Adaptation to Urgent Events:

UAVs dynamically re-route trajectories and alter altitudes to prioritize critical clusters immediately upon activation ($t = 200, 400, 600$).

Standard Service Recovery:

Once emergency waves clear, UAVs naturally return to tracking the center of mass of Normal users.

Results (4): Statistical Benchmarking

Objective: Maximize cumulative priority-weighted QoE: $\sum_t \sum_n w_{n,t} S_{n,t}$

Table: 10-Seed Performance Comparison Under Dynamic 3D Urban Environment (Mean ± Std. Dev.)

Method	Weighted QoE	Delivered MOS (Quality)			Wait Time (Latency, steps)		Coverage (MOS ≥ 1, %)		
	Utility (Points)	Emergency	Important	Normal	Emergency	Important	Emergency	Important	Normal
Static Centroid (Baseline)	33037.2 ± 4472.3	1.30 ± 0.87	1.59 ± 0.96	1.49 ± 0.11	122.57 ± 95.33	86.02 ± 52.43	$47.2\% \pm 31.4\%$	$48.9\% \pm 13.8\%$	$65.1\% \pm 3.1\%$
Dynamic K-Means (Baseline)	44284.9 ± 1977.2	4.39 ± 0.36	4.13 ± 0.69	1.30 ± 0.10	9.45 ± 7.69	7.70 ± 8.09	$94.5\% \pm 4.2\%$	$94.3\% \pm 6.7\%$	$53.7\% \pm 2.6\%$
Priority-Agnostic ($\alpha_{n,t} = 1.0$, $w_{n,t} = 0.1$)	33012.2 ± 3658.1	1.20 ± 0.75	1.41 ± 1.05	1.73 ± 0.12	64.58 ± 63.81	44.73 ± 53.15	$53.9\% \pm 30.3\%$	$61.5\% \pm 32.4\%$	$71.4\% \pm 3.0\%$
Proposed K-MAPPO	$\mathbf{45024.4 \pm 2482.0}$	$\mathbf{3.41 \pm 0.66}$	$\mathbf{3.54 \pm 1.01}$	$\mathbf{1.58 \pm 0.09}$	$\mathbf{7.38 \pm 4.97}$	$\mathbf{9.50 \pm 20.54}$	$\mathbf{92.3\% \pm 7.2\%}$	$\mathbf{91.7\% \pm 13.9\%}$	$\mathbf{65.7\% \pm 4.0\%}$

Key Findings:

- **+36% Global Utility Improvement:** K-MAPPO achieves 45,024 points vs. 33,037 (Static Centroid) and 33,012 (Priority-Agnostic).
- **94% Reduction in Emergency Latency:** Cuts average Emergency wait time down to 7.38 steps (vs. 122.57 steps for Static Centroid).
- **High Emergency Coverage:** Delivers 92.3% coverage and an average MOS of 3.41 for critical emergency nodes.
- **Maintains Normal user coverage at 65.7%** (preventing the steep drop to 53.7% seen in Dynamic K-Means).

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Conclusion

Key Thesis Contributions:

- **Urgency-Aware QoE Framework:** Shifted optimization focus from raw throughput to human-perceived MOS satisfaction scaled by urgency factors.
- **Two-Stage K-MAPPO Control:** Integrated macro K-Means initialization with decentralized MAPPO trajectory pathfinding under 3D urban terrain blockages.
- **Proven Fleet Performance:** Demonstrated +36% utility gains and a 94% latency drop for high-priority emergency users.

Future Research Horizons:

- **UAV Energy Constraints:** Incorporate battery capacity, propulsion power consumption, and flight dynamics to optimize long-term operational longevity.
- **Adaptive Resource Allocation**: Jointly optimize 3D trajectories with dynamic bandwidth and power distribution.

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Q&A